



DESIGN AND IMPLEMENTATION OF A BLOCKCHAIN-BASED SECURE IDENTITY AUTHENTICATION SYSTEM

PALLAB MANDAL (2022ITB001)
HARSHIT KUMAR (2022ITB011)
SUBHRANIL NANDY (2022ITB012)

UNDER THE GUIDANCE OF
DR. INDRAJIT BANERJEE

×

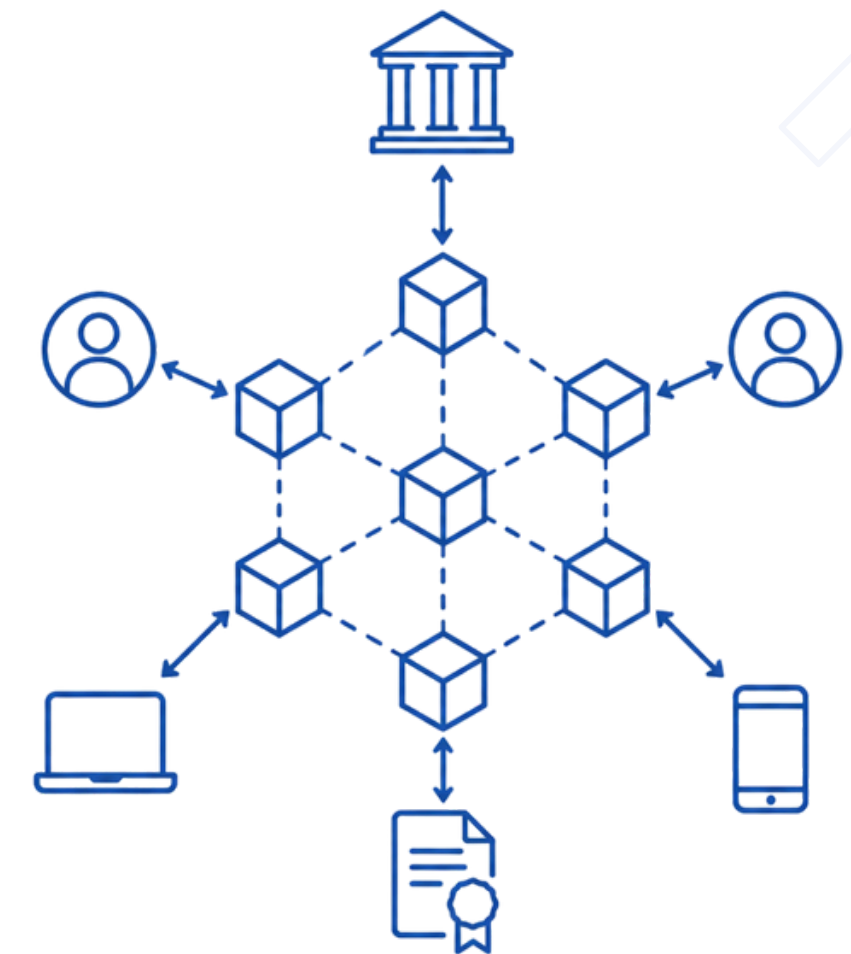
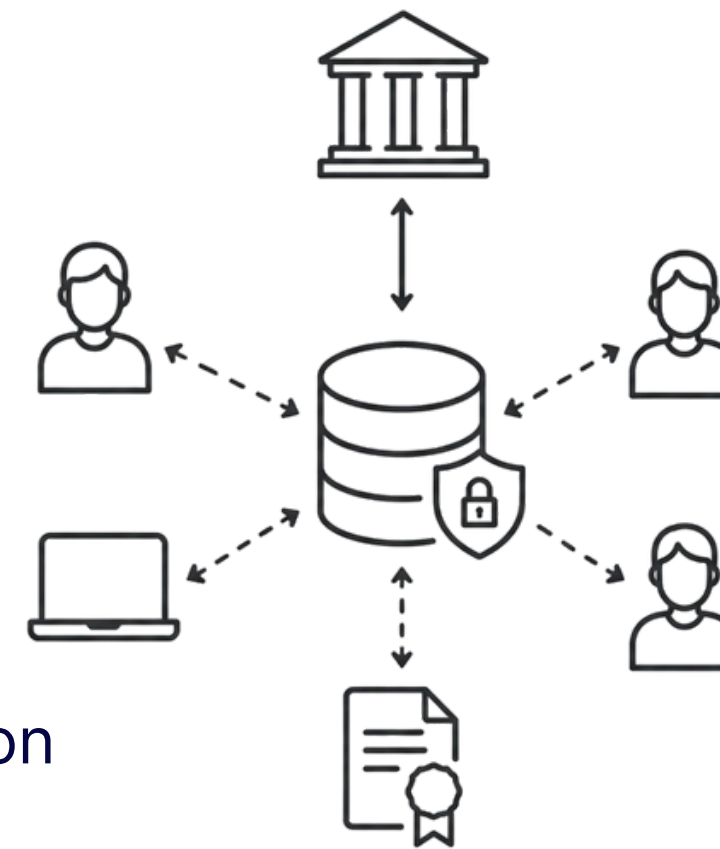
×

×

×

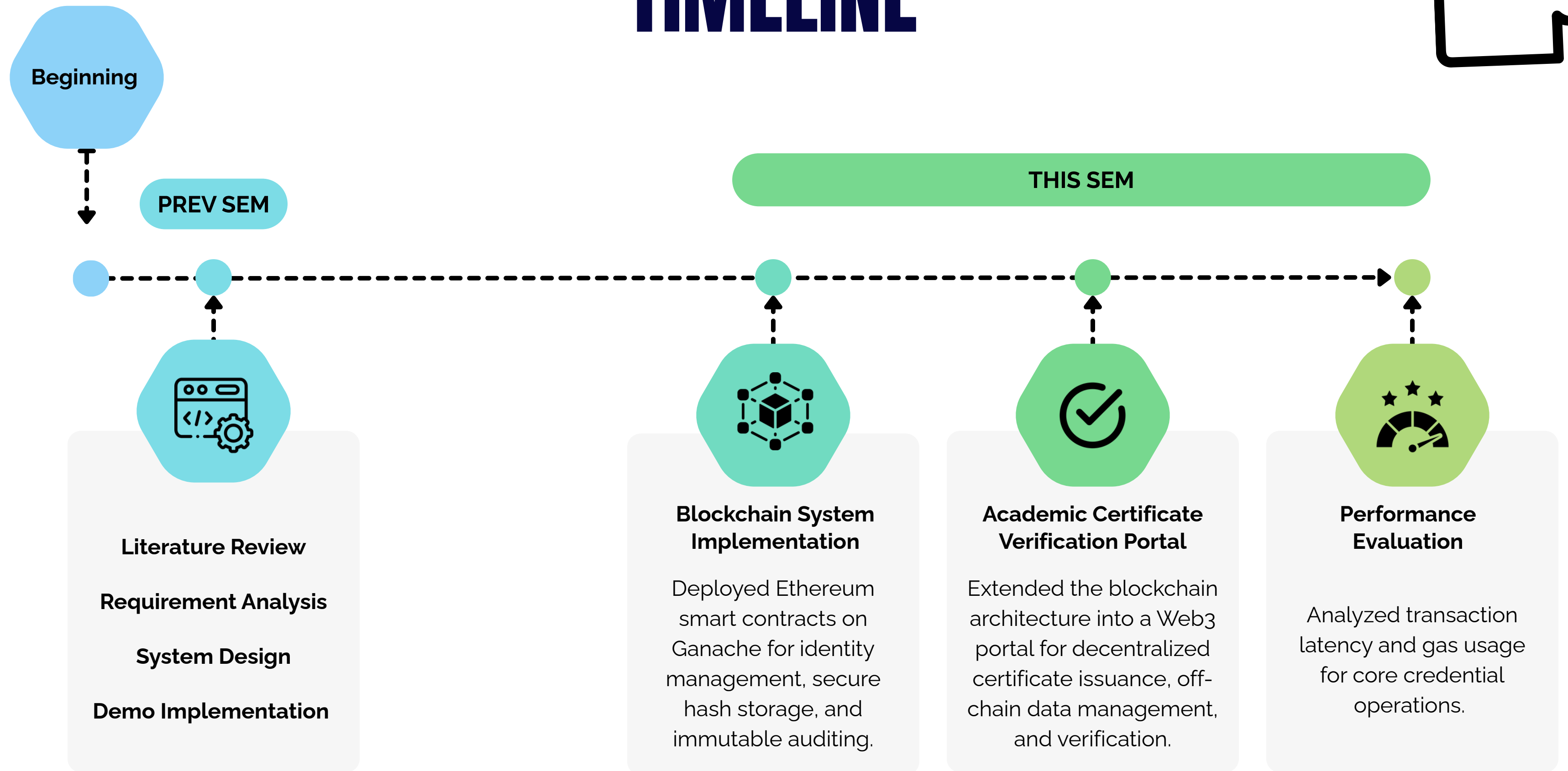
INTRODUCTION

- The Problem (Traditional Systems):
 - High verification latencies
 - Simple digital forgery
 - Grades and credentials vulnerable to unauthorized alterations
 - Centralized database vulnerabilities - **single points of failure**
- The Proposed Solution:
 - Web3-native identity mapping and credential verification portal
 - Leverages Ethereum smart contracts for tamper-resistant tracking
 - Implements a Self-Sovereign Identity (SSI) framework using Decentralized Identifiers (DIDs)

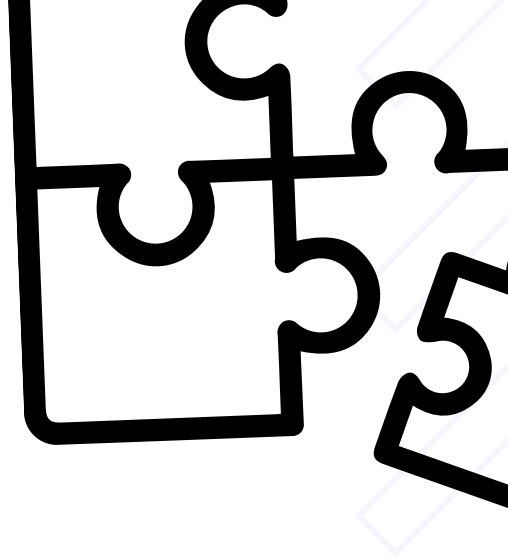


× × × ×

TIMELINE



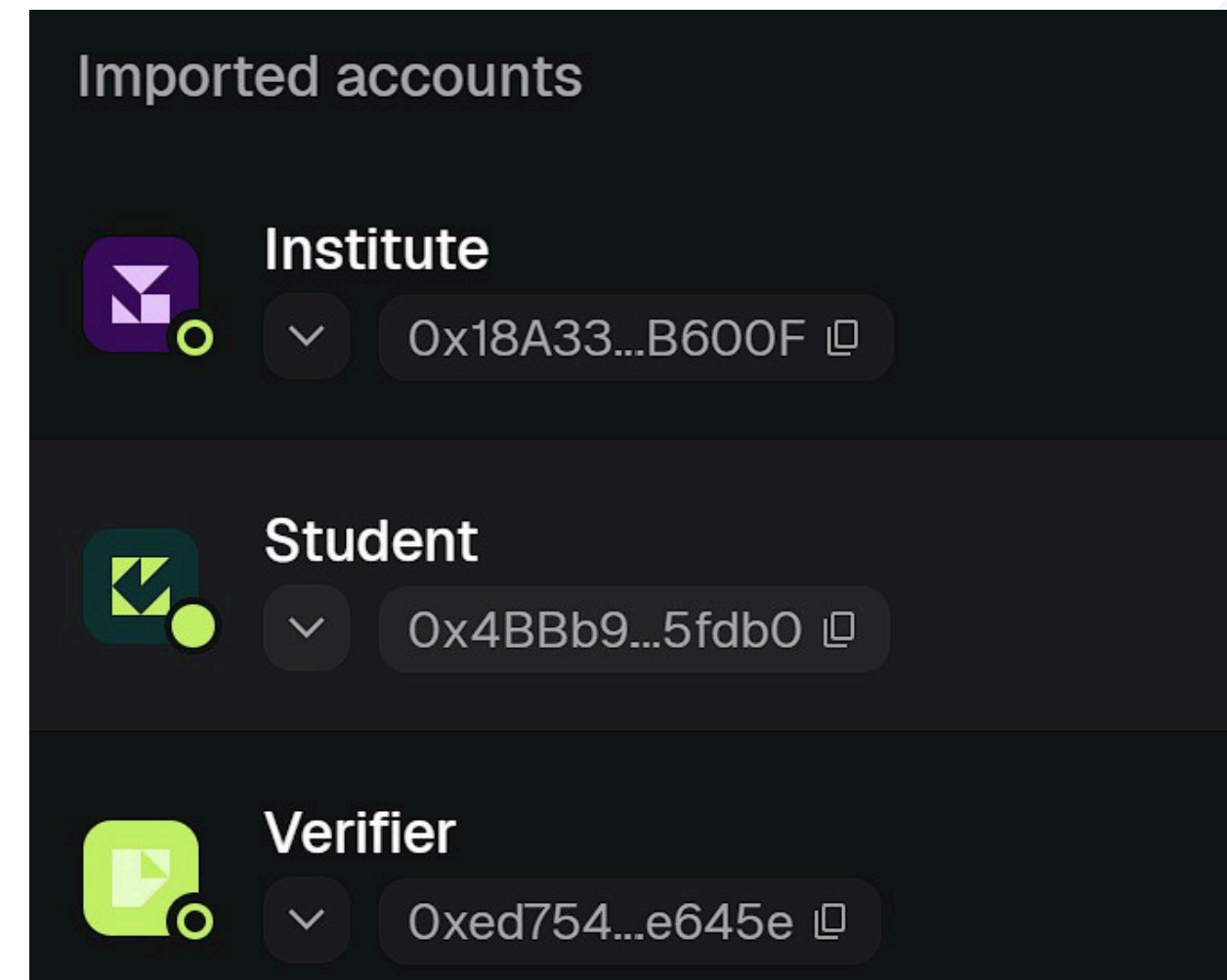
SYSTEM COMPONENTS & USERS



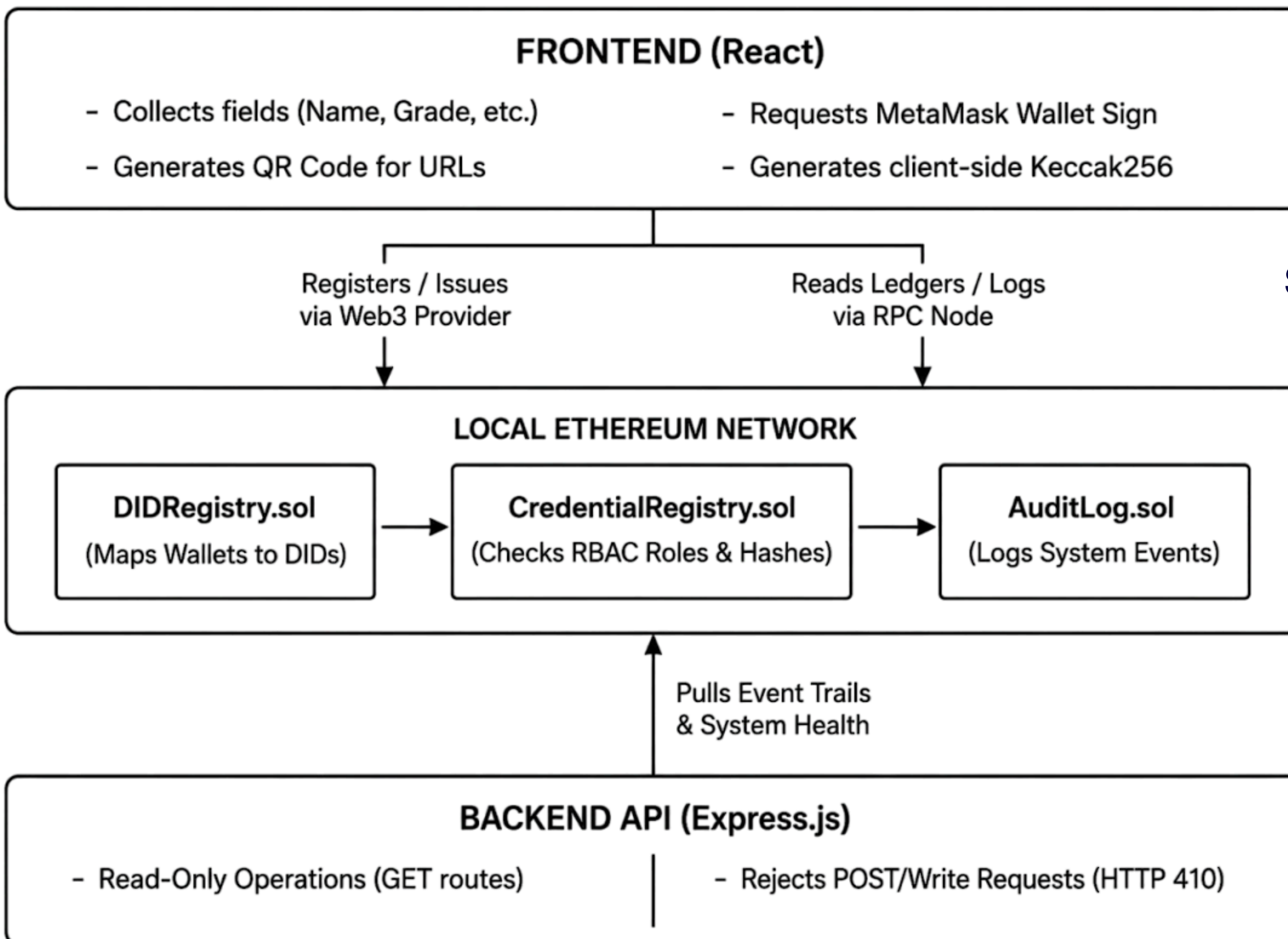
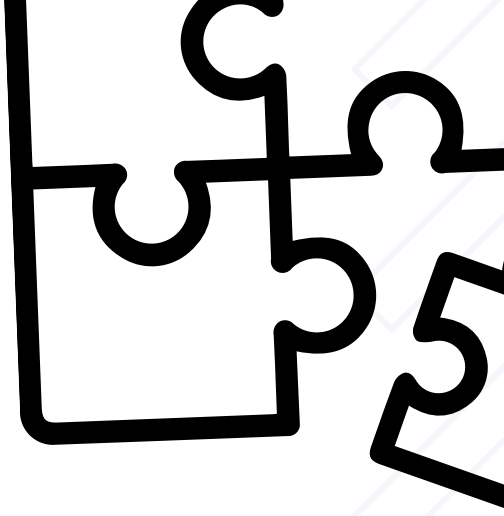
Core Components:

- Frontend (React): Web interface for
 - DID registration
 - Credential issuance, and
 - Instant verification.
- Backend (Express): Read-only data pipeline handling
 - API health check and
 - Audit trail queries.
- Smart Contracts (Solidity): Managing on-chain access control and tracking
 - DIDRegistry
 - CredentialRegistry
 - AuditLog
- Blockchain Network (Ganache): Local EVM ledger providing
 - Immutable storage
 - Tamper resistance.

× × × ×



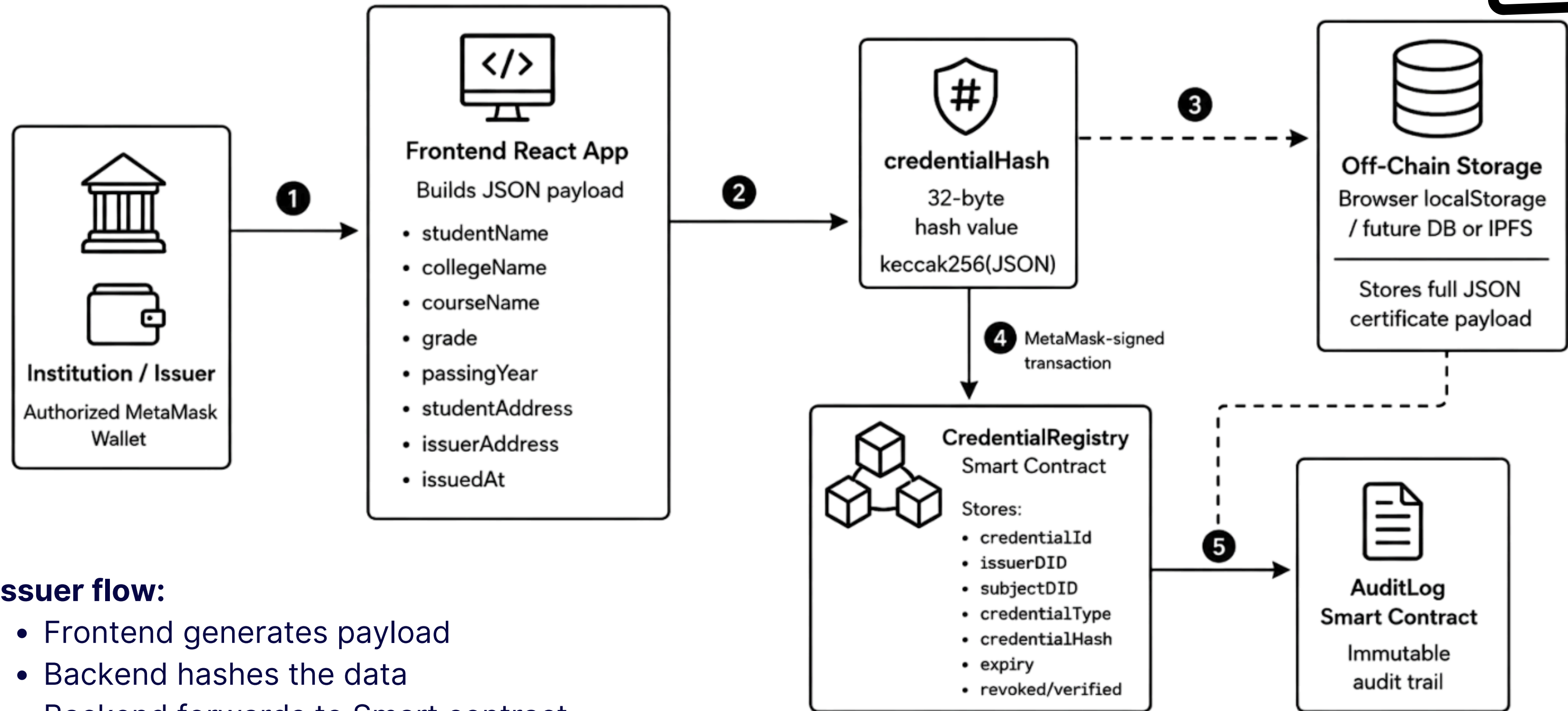
PROPOSED SYSTEM



Smart Contracts:

- DID Registry: Registers and maps users' decentralized identities to their wallet addresses.
- Credential Registry: Issues, stores, verifies, and revokes academic certificate credentials using on chain hashes.
- Audit Log: Records important actions like DID creation, certificate issuance, verification, and revocation immutably on the blockchain.

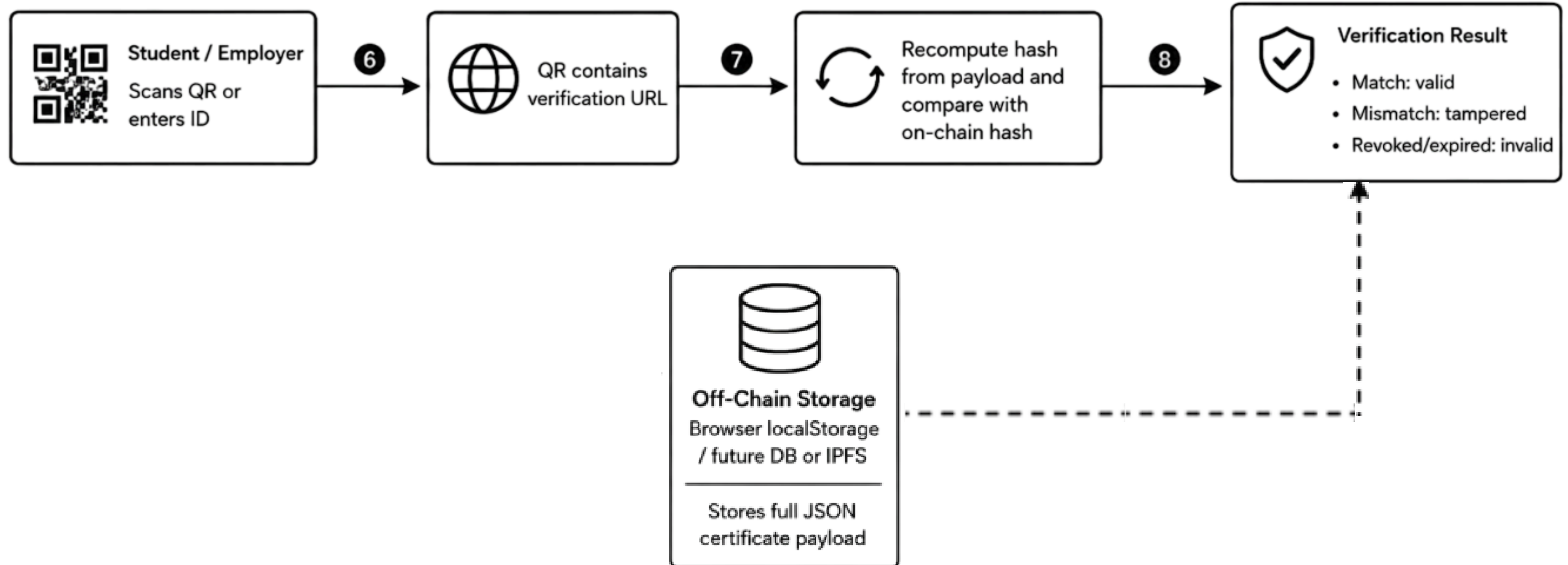
WORKFLOW



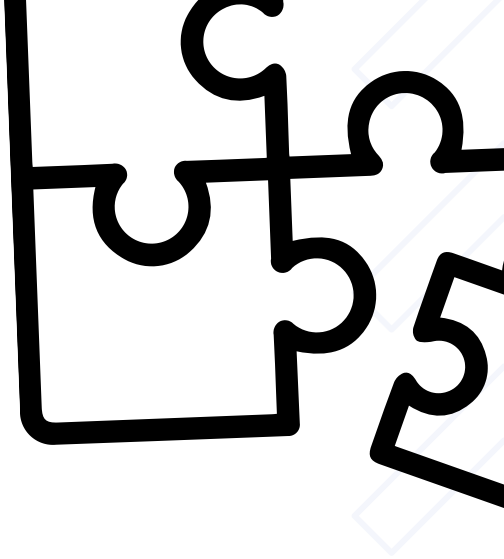
Issuer flow:

- Frontend generates payload
- Backend hashes the data
- Backend forwards to Smart contract
- All transactions are signed by user wallet(MetaMask)

VERIFICATION FLOW



PERFORMANCE & RESULT



Write Transaction Metrics

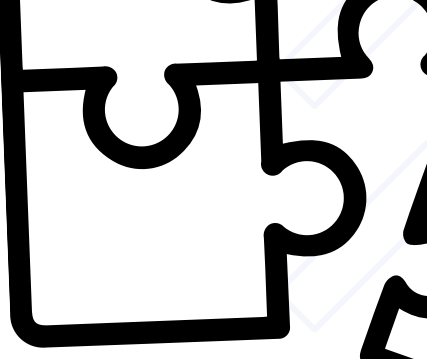
Operation	Average Latency	Average Gas Used
DID Registration	268.84 ms	216,891
Credential Issue	420.02 ms	432,400
Credential Verification	380.38 ms	304,101

- Blockchain writes are slower due to consensus validation and EVM execution overhead.
- Gas usage remains efficient because only the certificate hash is stored on chain.
- Backend read APIs are comparable with other Express applications.
- Global audit log retrieval is slower because historical blockchain events must be scanned and assembled.

Backend Read Latency Metrics

Backend Endpoint	Average Latency
GET /health	48.05 ms
GET /did/:didHash	48.74 ms
GET /credential/:credentialId	44.81 ms
GET /audit/did/:didHash/ids	35.54 ms
GET /audit/did/:didHash	151.48 ms
GET /audit/credential/:credentialId	147.86 ms
GET /audit/recent/:limit	384.10 ms

PERFORMANCE & RESULT



SYSTEM / CERTIFICATE VERIFICATION

Academic Certificate Verification Portal

ACCOUNT:

0x4bbb965e...fdb0

BALANCE:

100 ETH

NETWORK ID:

1337

PHASE 1: REGISTRATION & ISSUANCE

PHASE 2: AUTHENTICATION & VERIFICATION

PHASE 3: ACCESS & AUDIT

Success: DID registered! Hash: 0x361132a4a54c1fca5937c986e4607f69e0f9399fa54ee59c37a12c21efc51549

PHASE 1: REGISTRATION & ISSUANCE

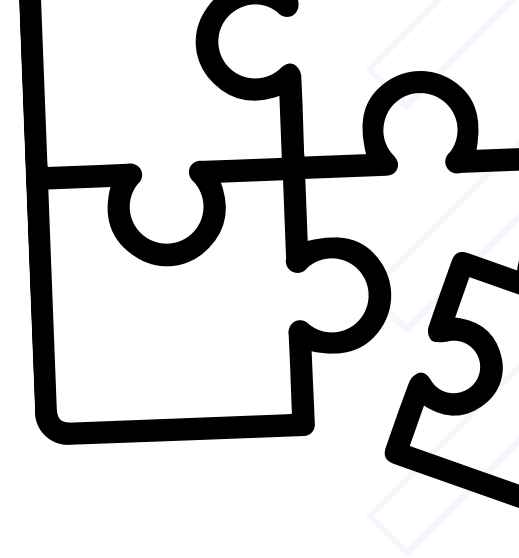
Register your Decentralized Identifier (DID) and create verifiable credentials

DID STRING (E.G., DID:BLOCKCHAIN:123ABC)

DID:2022ITB001

REGISTER DID

PERFORMANCE & RESULT



SYSTEM / CERTIFICATE VERIFICATION

Academic Certificate Verification Portal

ACCOUNT:	BALANCE:	NETWORK ID:
0x18a33bc7...600f	99.99 ETH	1337

PHASE 1: REGISTRATION & ISSUANCE

PHASE 2: AUTHENTICATION & VERIFICATION

PHASE 3: ACCESS & AUDIT

Success: DID registered! Hash: 0x4bbacc0f52dc429c6fc0d71390fc1c524f3ebbb2a5bb7121c44b72b811fc6871

PHASE 2: AUTHENTICATION & VERIFICATION

Issue, verify, and audit academic certificates using on-chain credential hashes

STUDENT WALLET ADDRESS

0x4BBb965EDCF61140Ac1D8Dfe610F5c1fcDB5fdb0

STUDENT NAME

Harshit

COLLEGE NAME

IIEST

COURSE

IT

GRADE

9

PASSING YEAR

2026

ISSUE ACADEMIC CERTIFICATE

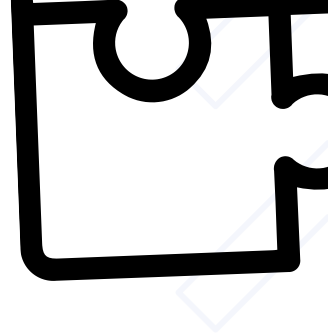
ISSUED CERTIFICATE

CREDENTIAL ID: 0x14955840fbcd25e883fdf2ce2729b0f430eb14e6dbe9debc5c8259526915ca9a



```
{
  "studentName": "Harshit",
  "collegeName": "IIEST",
  "courseName": "IT",
  "grade": "9",
  "passingYear": "2026",
  "studentAddress": "0x4BBb965EDCF61140Ac1D8Dfe610F5c1fcDB5fdb0",
  "issuerAddress": "0x18A33bc79d5b1e45b08B56b7e9643815019B600F",
  "issuedAt": "2026-05-19T12:22:03.710Z"
}
```

PERFORMANCE & RESULT



CREDENTIAL ID OR QR URL

0x14955840fbcd25e883fdf2ce2729b0f430eb14e6dbe9debc5c8259526915ca9a

VERIFY & CHECK INTEGRITY

VERIFIED CREDENTIAL JSON

```
{
  "credentialId": "0x14955840fbcd25e883fdf2ce2729b0f430eb14e6dbe9debc5c8259526915ca9a",
  "issuerDID": "0x4bbacc0f52dc429c6fc0d71390fc1c524f3ebbb2a5bb7121c44b72b811fc6871",
  "subjectDID": "0x361132a4a54c1fca5937c986e4607f69e0f9399fa54ee59c37a12c21efc51549",
  "credentialType": "ACADEMIC_CERTIFICATE",
  "credentialHash": "0xc12a7258383c345489bca19b9ef89233bbb5c69b74da6aec5ece955d4d83e3e2",
  "issuedAt": "1779193334",
  "expiresAt": "2094553334",
  "isRevoked": false,
  "isVerified": true,
  "integrityVerified": true,
  "payload": {
    "studentName": "Harshit",
    "collegeName": "IIEST",
    "courseName": "IT",
    "grade": "9",
    "passingYear": "2026",
    "studentAddress": "0x4BBb965EDCF61140Ac1D8Dfe610F5c1fcDB5fdb0",
    "issuerAddress": "0x18A33bc79d5b1e45b08B56b7e9643815019B600F",
    "issuedAt": "2026-05-19T12:22:03.710Z"
  }
}
```

DID HASH

0x361132a4a54c1fca5937c986e4607f69e0f9399fa54ee59c37a12c21efc51549

- DID AUDIT IDS
- DID AUDIT RECORDS
- RECENT RECORDS

DID AUDIT IDS (MINIMAL PATH)

```
{
  "success": true,
  "didHash": "0x361132a4a54c1fca5937c986e4607f69e0f9399fa54ee59c37a12c21efc51549",
  "auditRecords": [
    1,
    3,
    4
  ]
}
```

DID AUDIT RECORDS (ENRICHED PATH)

RECORD ID	ACTION	DID STRING	ACTOR	SUBJECT DID	RELATED ENTITY	DETAILS
1	DID_CREATED	DID:2022ITB001	0x4BBb965EDC...cDB5fdb0	0x361132a4a5...efc51549	0x361132a4a5...efc51549	DID created
3	CREDENTIAL_ISSUED	DID:2022ITB001	0x6493D3Ae3F...79a10C57	0x361132a4a5...efc51549	0x14955840fb...6915ca9a	Credential issued
4	CREDENTIAL_VERIFIED	DID:2022ITB001	0x6493D3Ae3F...79a10C57	0x361132a4a5...efc51549	0x14955840fb...6915ca9a	Credential verified an...

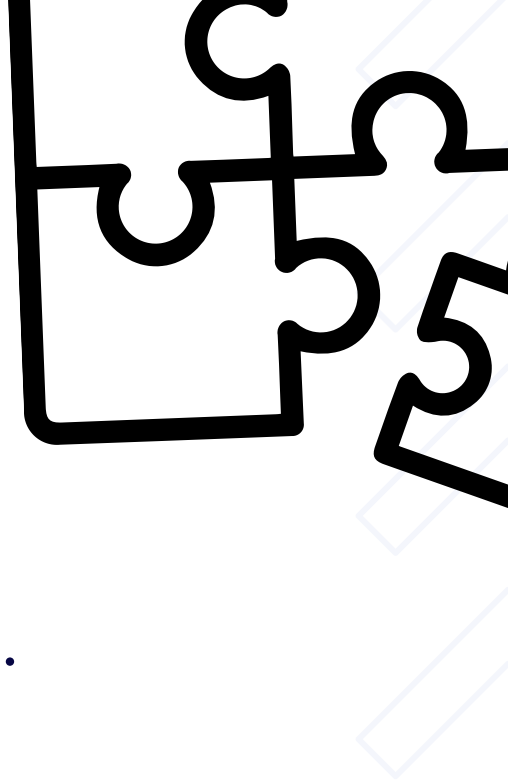
RECENT AUDIT RECORDS

RECORD ID	ACTION	DID STRING	ACTOR	SUBJECT DID	RELATED ENTITY	DETAILS
-----------	--------	------------	-------	-------------	----------------	---------

APPLICATION

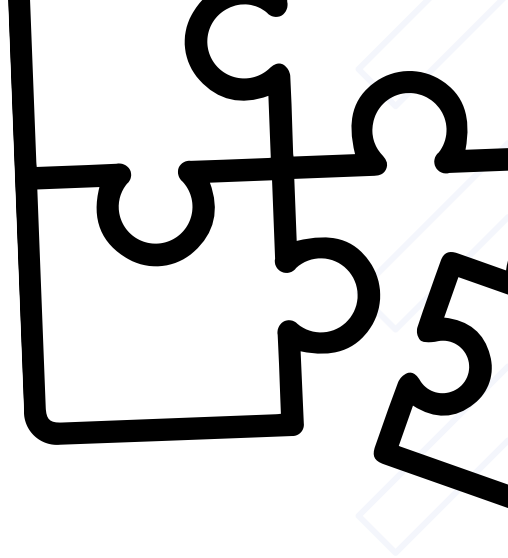
- Corporate Credentials: Issue secure work experience letters.
 - Modification: Change payload to job details; restrict issuer access to HR only.
- Event Ticketing: Prevent counterfeit tickets and unauthorized access.
 - Modification: Update schema for seat/date; add transfer logic for ticket resale.
- Supply Chain: Track authenticity and origin of physical goods.
 - Modification: Map DIDs to suppliers; track batch numbers off-chain.
- Healthcare Records: Issue tamper-proof medical prescriptions.
 - Modification: Shift payload to medical data; mandate ZKPs for patient privacy.
- Government IDs: Verify digital licenses and permits.
 - Modification: Grant contract authority to state bodies; handle expiration renewals.
- Intellectual Property: Prove digital ownership for art or code.
 - Modification: Allow creator self-issuance; map the credential to the original file's hash.

LIMITATION & FUTURE SCOPE



- IPFS Integration: Store encrypted payloads off chain using CIDs for true decentralized availability.
- Zero-Knowledge Proofs (ZKPs): Enable selective data disclosure to ensure privacy and GDPR compliance.
- Mainnet & Layer-2 Deployment: Migrate to Ethereum Mainnet using Polygon/Arbitrum for fast, low-cost transactions.
- Institutional Interoperability: Adopt W3C standards and build a multi-university consortium dashboard.
- Production Readiness: Strengthen decentralized identity and audit transparency for enterprise-scale scaling.

REFERENCES



[1] Research Paper 1 (Survey)

W. Chan, K. Gai, J. Yu, and L. Zhu, "Blockchain-Assisted Self-Sovereign Identities on Education: A Survey," Blockchains, vol. 3, no. 1, p. 3, 2025. Available: <https://doi.org/10.3390/blockchains3010003>

[2] Research Paper 2 (Micro-accreditation)

A. Mikroyannidis, A. Third, and J. Domingue, "Blockchain-Based Decentralised Micro-Accreditation for Lifelong Learning," Interactive Learning Environments, vol. 33, no. 3, pp. 2201–2215, 2025. doi: 10.1080/10494820.2024.2401485.

[3] Official W3C Documentation (Web Standards)

M. Sporny, D. Longley, and D. Chadwick, "Verifiable Credentials Data Model v1.1," W3C Recommendation, World Wide Web Consortium, 2022. [Online]. Available: <https://www.w3.org/TR/vc-data-model/>

[4] Canva Slide Template

× × × ×

THANK YOU